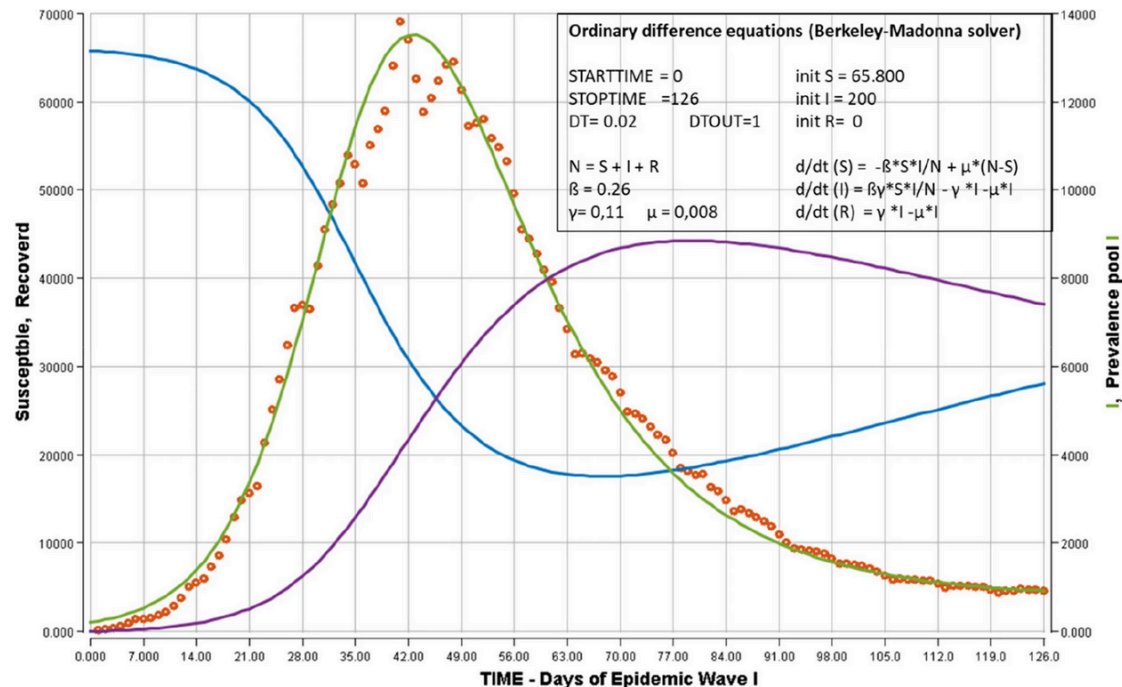


# Foundations of Data Science

Lecture 1: Build, compute, critique, repeat

Prof. Gilles Louppe  
[g.louppe@uliege.be](mailto:g.louppe@uliege.be)

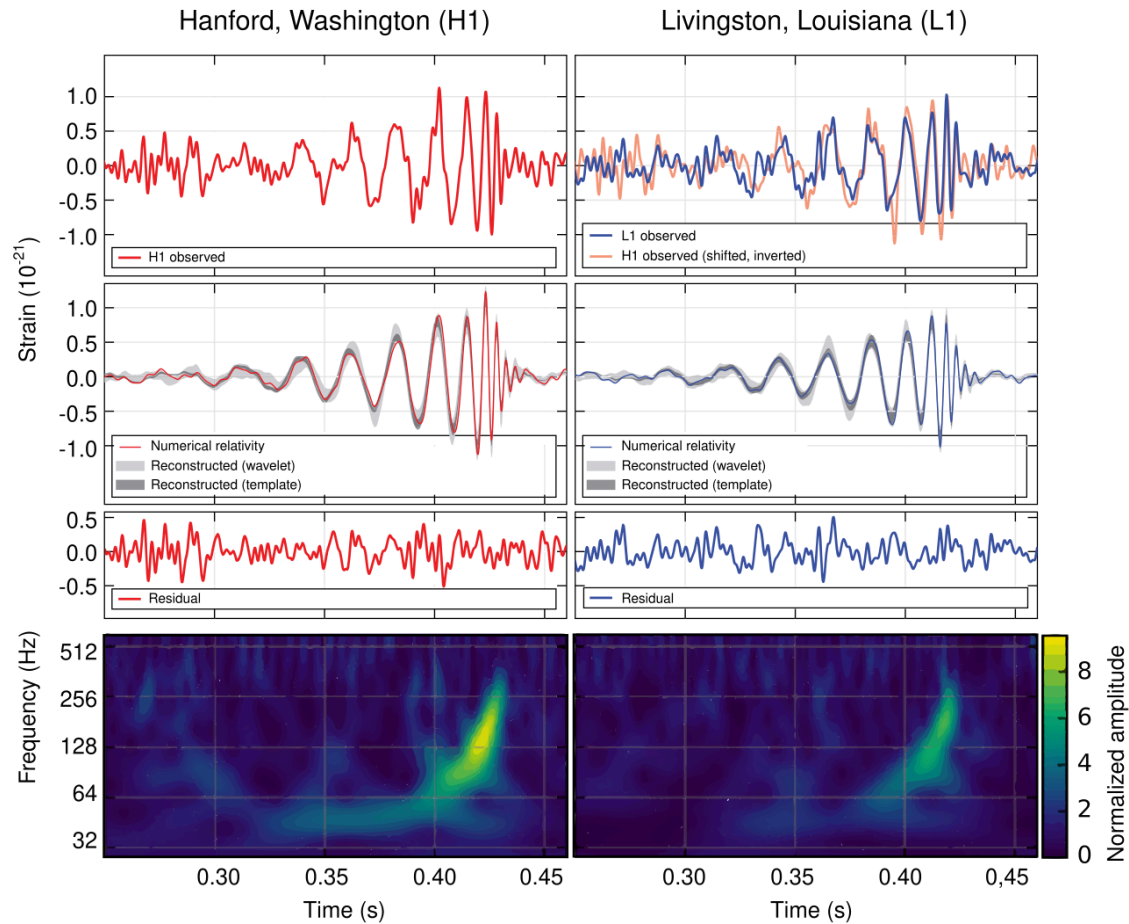
# COVID-19 epidemiological models



Open SIR model fitted to observations of the first Covid-19 epidemic wave – Belgium. The evolution of daily prevalence pool I was estimated from the daily incidence of reported cases (red circles). These observed values were fitted by ordinary differential equations (ODE) to SIR model. Left vertical axis: modeled Susceptible compartment (blue); modeled Recovered compartment (violet). Right axis: modeled daily prevalence pool I compartment. The insert contains the equations of ODE in Berkeley-Madonna language, the initial values of S, I and R compartments, the  $\beta$  and  $\gamma$  transition parameters. DT signifies that the fitting of model to observations with ODE was operated for each  $\Delta$  time = 0,02 days. DTOUT indicates the  $\Delta$  time for output printing

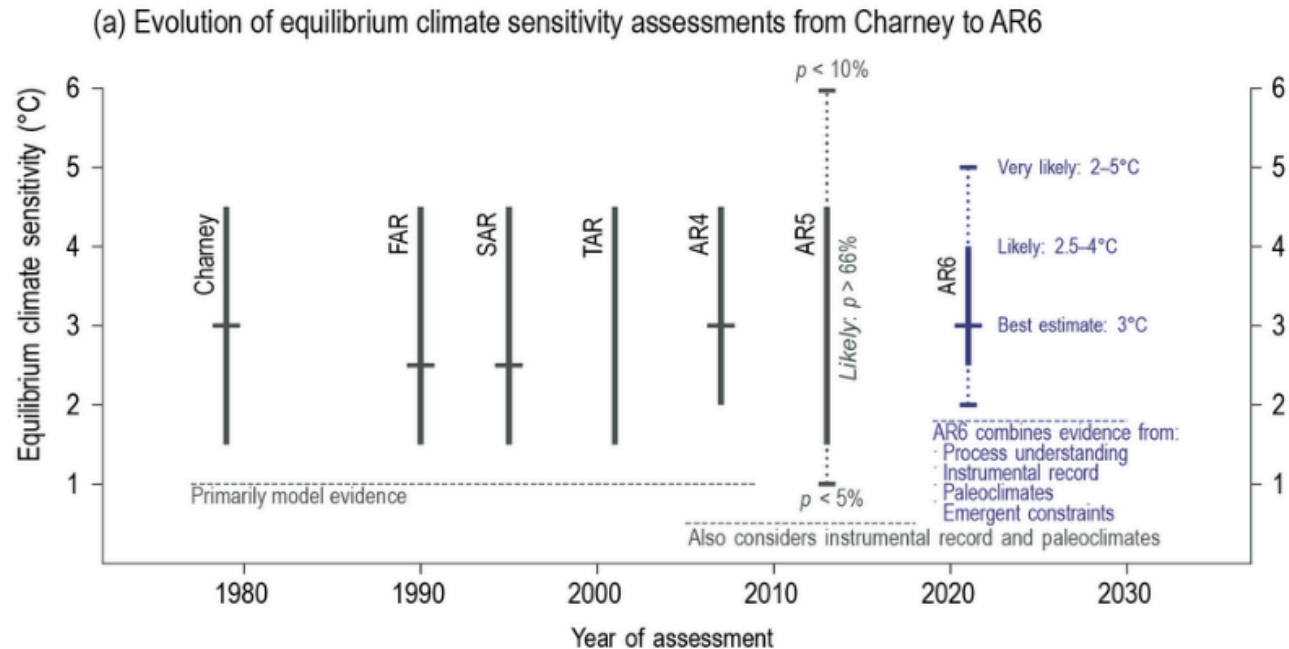
Predictions of the first wave of COVID-19 in Belgium, 2020.  
Many models failed to predict the peak and duration of the wave.

# Gravitational wave detection



First direct detection of gravitational waves, LIGO, 2015.  
Many false alarms before the first confirmed detection.

# Equilibrium climate sensitivity



What is the Earth's temperature increase if we double atmospheric CO<sub>2</sub>?  
Over 40 years, uncertainty narrowed from 3°C range to 1.5°C range.



# Box's loop

All models are wrong, but some are useful. -- George Box.



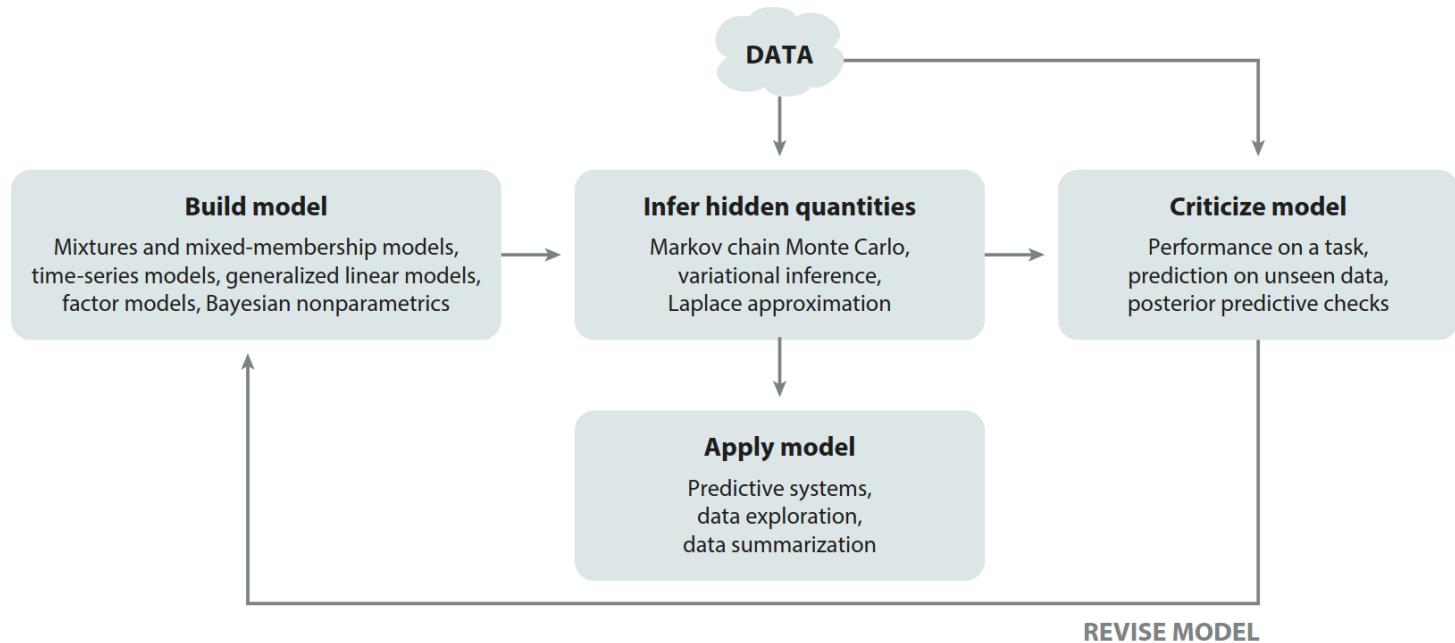
## What is data science?

Data science is the discipline of **extracting knowledge** from data through the iterative application of the **scientific method**.

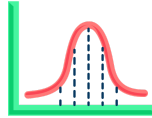
But how?



## Box's loop



Scientific inquiry as an iterative process: build, compute, critique, repeat.



## Step 1: Build

The first step to understanding a phenomenon is to **build a model** of it, as a simplified representation that captures its essential aspects.

- A model specifies assumptions about the data generating process.
- It encodes domain knowledge and constraints.
- It is formulated within an appropriate mathematical abstraction.
- It defines what you observe and what you do not observe but assume exists.



## Step 2: Compute

The next step is to **compute** what the data tells you about the phenomenon of interest under the assumptions of your model.

- Fitting the model to data involves solving an optimization problem.
- Inference is used to answer questions about unobserved quantities.
- Prediction is used to answer questions about future or unseen data.



## Step 3: Critique

The third step is to **critique** the model and its predictions, to assess whether they are consistent with the data and domain knowledge.

- Compare predictions to observed data.
- Identify model limitations and mismatches.
- Reject the model if it fails to capture key aspects of the data.

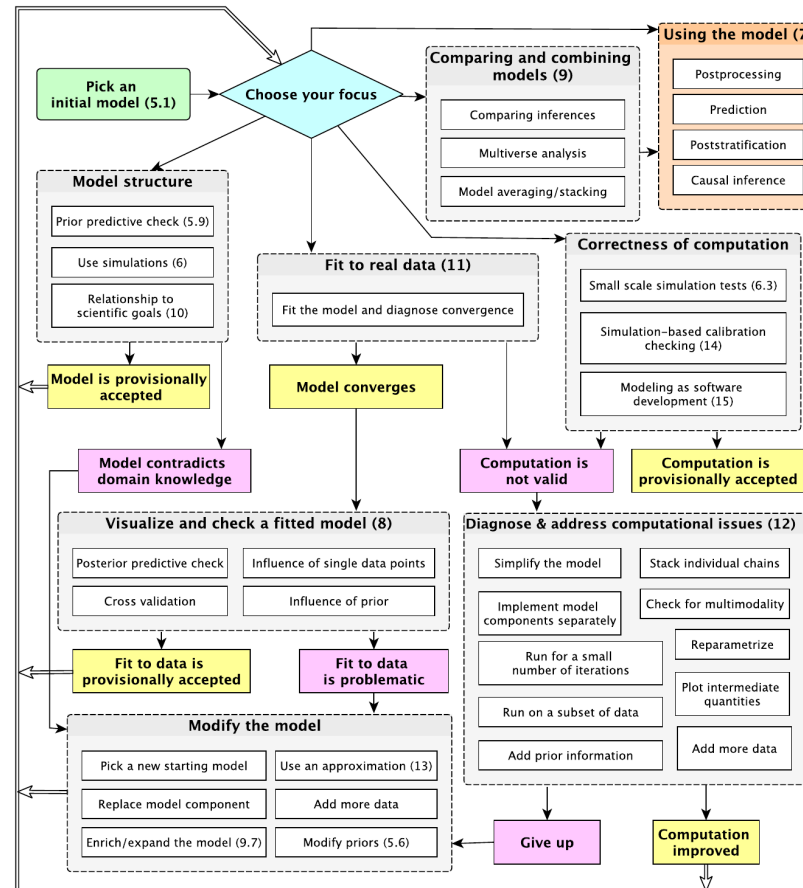


## Step 4: Repeat

What you learn from the critique step informs how to **repeat** the process.

- Add complexity to the model to address its shortcomings.
- Simplify the model to improve interpretability.
- Change the model to explore alternative hypotheses.

# The loop, in practice

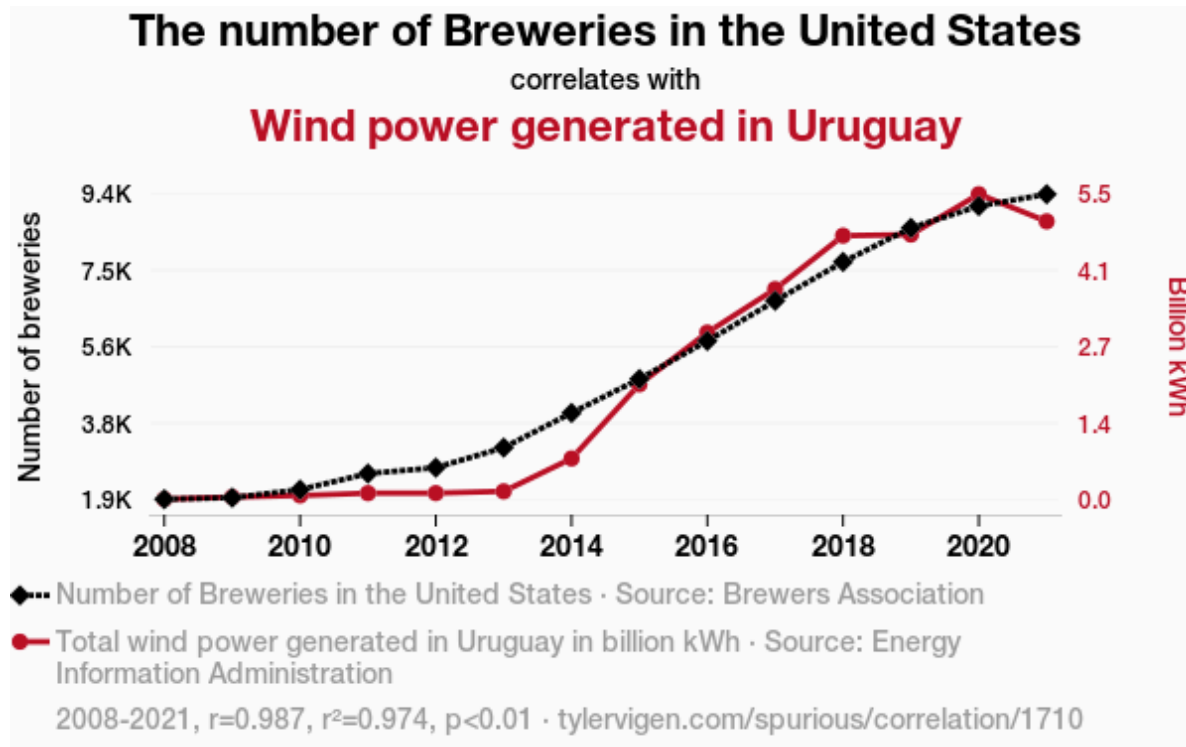


Bayesian workflow: the steps and paths an analysis may go through.

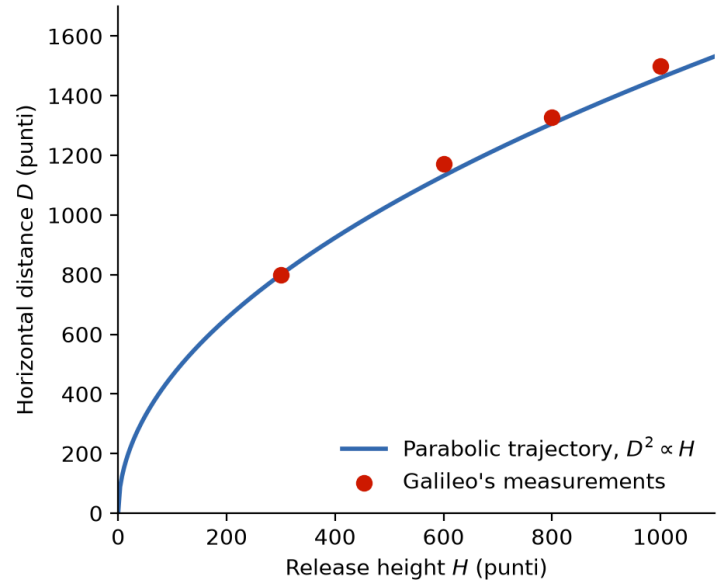
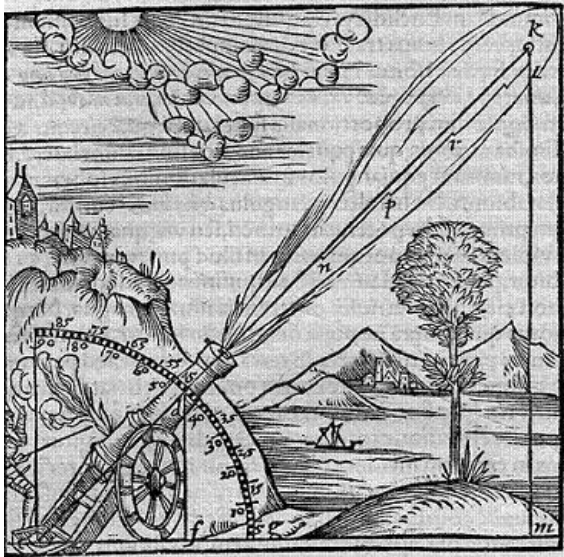
## Why this approach matters?

The Fourth paradigm (Hey et al, 2009) of science emphasizes the importance of data-intensive scientific discovery. However, data alone is not enough.

- Data without theory leads to spurious correlations.
- Theory without data leads to ungrounded speculation.
- Together, they enable **robust scientific inquiry**.



Spurious correlation can easily mislead data analysis.



Theory without data: the trajectory implied by Aristotelian physics (left) against Galileo's measurements, consistent with a parabola (right).

The scientific method, as embodied in Box's loop, provides a principled framework for data analysis. It contrasts with **ad-hoc data analysis practices that often lead to unreliable results**.

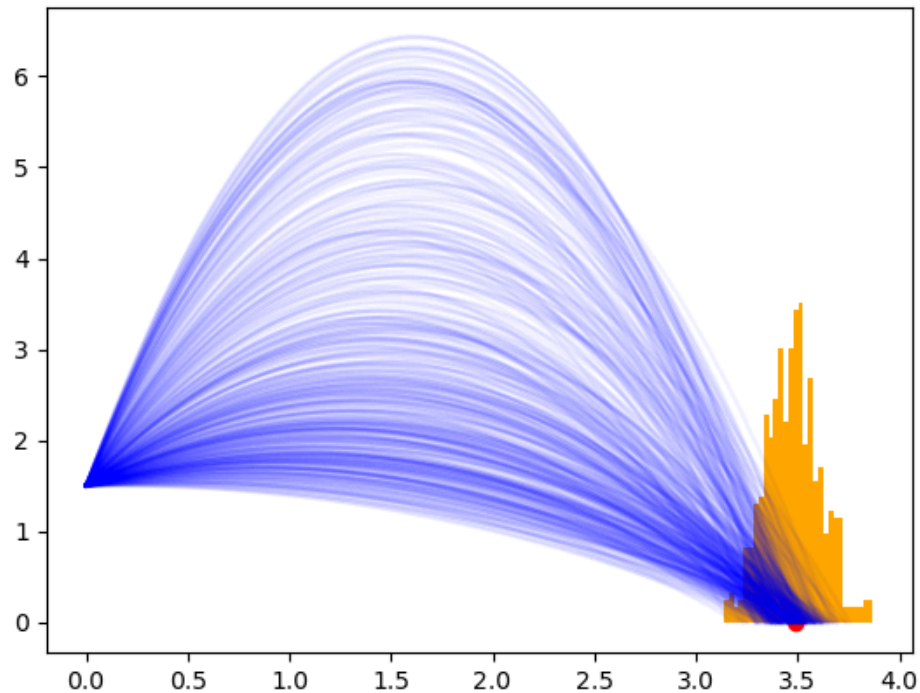
- Throw algorithms at data and see what sticks.
- Focus only on prediction accuracy.
- Treat models as black boxes.
- Stop at the first somewhat satisfactory result.

Box's loop encourages

- transparent reasoning about assumptions,
- understanding mechanisms behind data, not just correlations,
- honest assessment of model limitations,
- continuous improvement and learning.

## Today's example: Projectile motion

A ball is thrown and lands at some measured distance  $x$ . What can infer about the initial velocity  $v$  and angle  $\alpha$  of the throw?



Let's code!

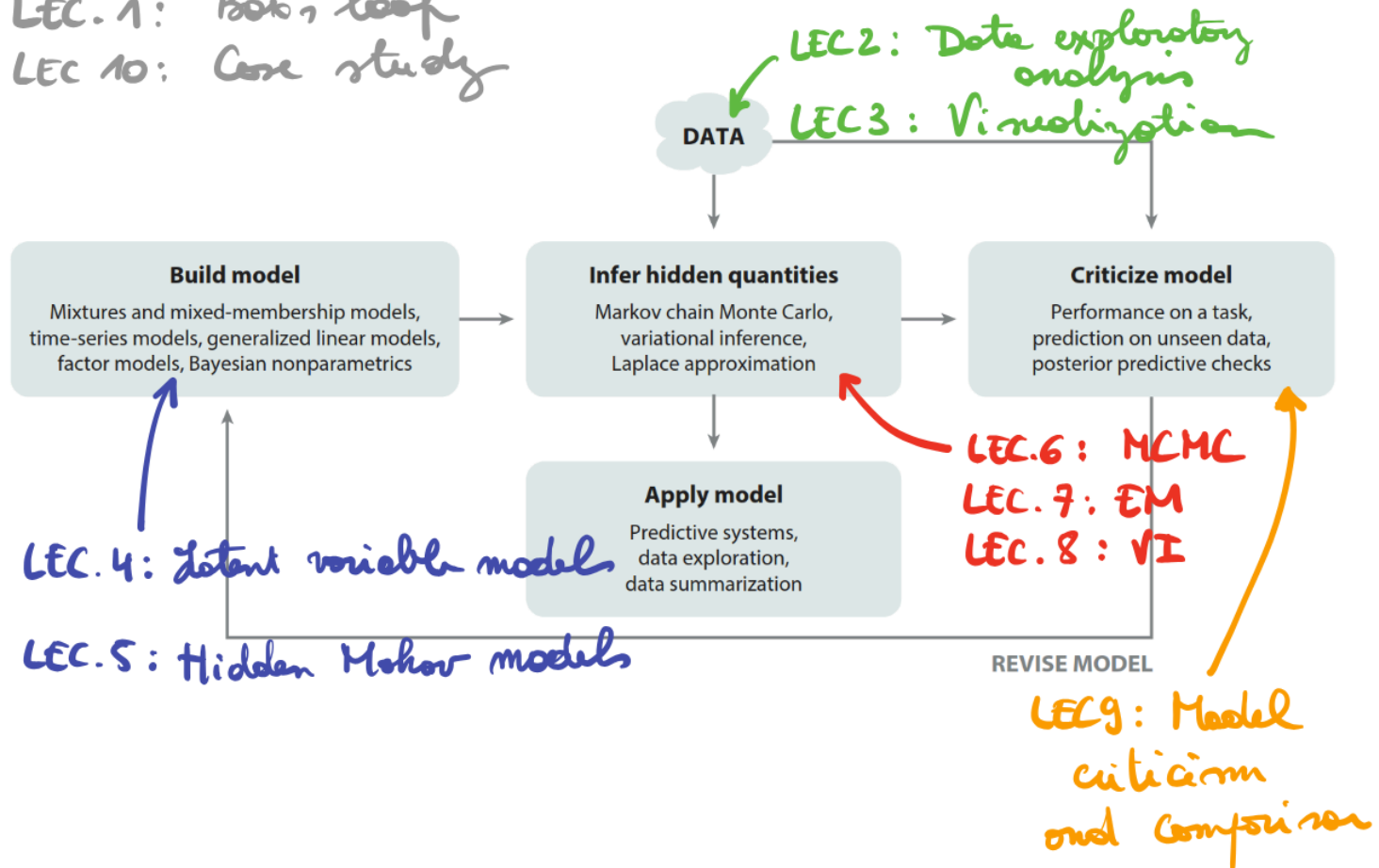
- Open a Jupyter notebook and follow along (or check `nb01.ipynb` after the lecture).
- Implement Box's loop step by step:
  1. Build a simple physical model of projectile motion.
  2. Compute estimates of  $\mathbf{v}$  and  $\alpha$  from data.
  3. Critique the model fit and predictions.
  4. Repeat by refining the model to account for more realistic factors.

**DATS0001**

# Outline

- Lecture 1: Build, compute, critique, repeat
- Lecture 2: Data and exploratory analysis
- Lecture 3: Visualization
- Lecture 4: Latent variable models
- Lecture 5: State-space models
- Lecture 6: Markov Chain Monte Carlo
- Lecture 7: Model criticism and validation
- Lecture 8: Expectation-Maximization
- Lecture 9: Variational Inference
- Lecture 10: Simulation-based inference
- Lecture 11: Case study

LEC. 1: Box's loop  
LEC 10: Core study



## **My mission**

Teach you how to think like a scientist in the age of data.

By the end of this course, you will be able to explore, model, and reason about data in a principled way, but also to communicate your findings effectively.

The end.